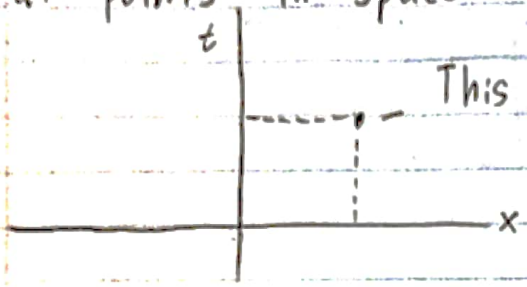
A review of Special Relativity (SR)

Recall how in physics we make measurements at points in space-time.



This point notions "something happens here", / "event" in SR

When hearing "event", think a point in space-time, the idea being we assign numbers (t,x) to events in space-time by setting up a reference frame.

Reference Frame:
Lay down some meter sticks & some clocks, and do something in space-time. Then look at the meters and clocks at that point and assign some numbers to that location and time of the event.

The key point of relativity is that the assigned numbers are completely man-made. They are artificial regardless of the numbers assigned to the event; it's an event - something happened here. It could be the case that in another frame of reference, different numbers are assigned to the event.

↳ RF 1 => (t, x)
↳ RF 2 => (t', x') } No reason at all for $(t,x) = (t',x')$. Their frame of ref. can be different.

We can ask "how are the coordinates of the event related? Is there a relationship between (t,x) & (t',x')?"

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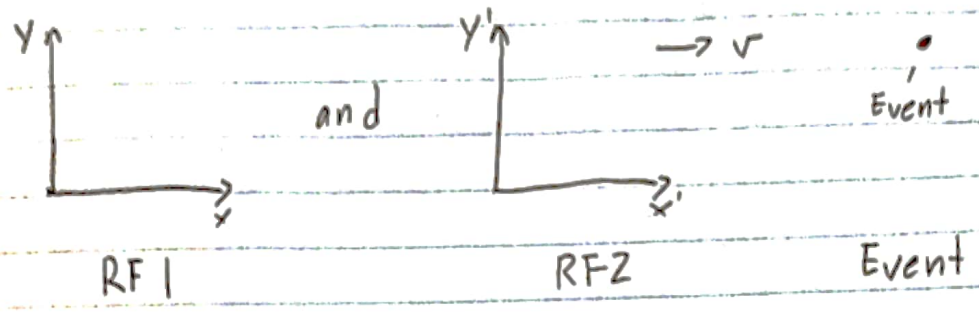
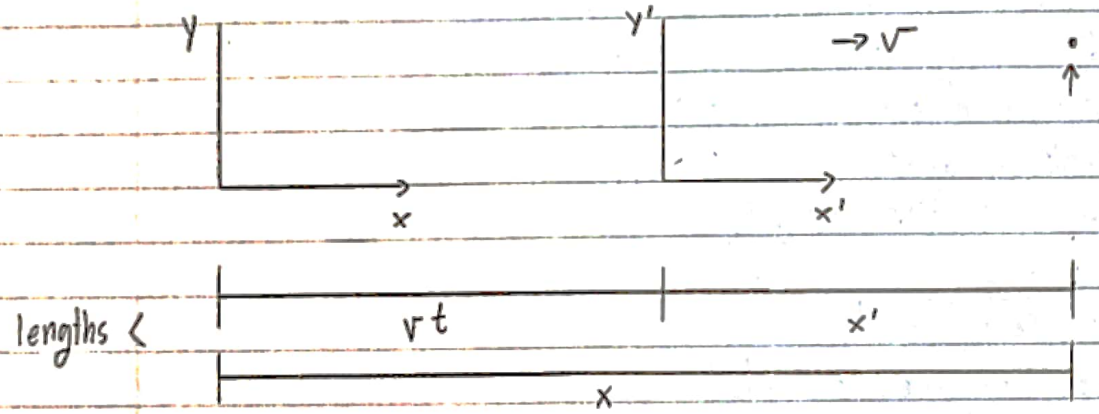
Galilean Relativity

Suppose we have 2 coord. systems in 2-D:



RF2 is moving at velocity v, relative to RF1, and we have some event that happens in space-time.

Ask "how can an observer relate the coordinates of the event using systems RF1 & RF2, assigned to the event?". The sol'n is as follows:



See how: $x' = x - vt$
$t' = t$

This is called a Galilean transformation and we are assuming both observations agree on the time coordinate.

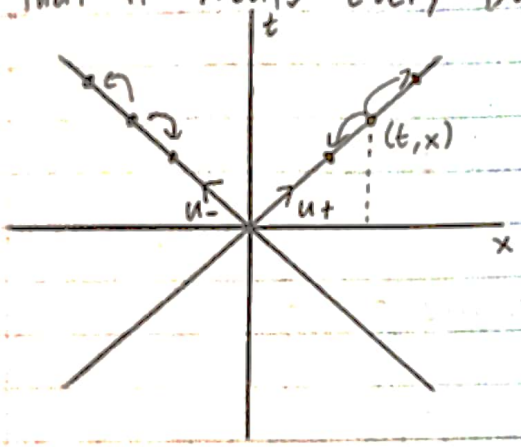
The problem with the Galilean transformation above is the conflict with relativity and how it doesn't keep the speed of light constant. c should be measured to be the same in both RF1 & RF2.

Say we shoot a laser-pulse and after some time it hits a mirror. That event will have coords:

$$(t, x=ct) \to (t', x' = ct - vt)$$
$$= (c-v)t$$
$$= (c-v)t'$$

We assume that as we move at v, we will see that the light travels at a reduced velocity. $(c-v)$ Therefore we need a new change of coordinates that keep c constant.

Assume a linear transformation (which can be argued that it treats every point in space-time the same).



We have 2 light rays moving in both $\pm x$ and we want to find a change in coords that keep the speed of light the same. This means the point, left, can only move on the line.

With the points only being able to move on the line, this means the lines are the span of eigenvectors of the Lorentz transformation.

Normalizing vectors u_+ & u_-:

$$u_+ = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \quad u_- = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$$

$\hookrightarrow \Lambda u_\pm = \lambda_\pm u_\pm$

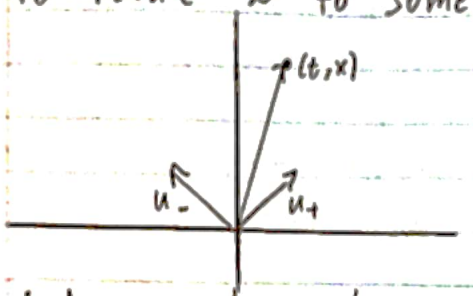
Matrix of Lorentz transformation in basis of $u_\pm$:

$$\Lambda_D = \begin{pmatrix}\lambda_+ & 0\\ 0 & \lambda_-\end{pmatrix}, \quad \Lambda_D^{-1} = \begin{pmatrix}\lambda_- & 0\\ 0 & \lambda_+\end{pmatrix} \Rightarrow \Lambda_D \Lambda_D^{-1} = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \Rightarrow \lambda_+\lambda_- = 1 \Rightarrow \therefore \lambda_\pm > 0$$

With $\lambda_+\lambda_- = 1$ & $\lambda_\pm > 0$ established, we are allowed to parameterize $\lambda_\pm$:

$$\lambda_\pm = e^{\mp\eta}, \quad \eta = 0 \text{ when } v = 0$$

At this stage we are done, we've determined the lorentz transformation and we just need to change basis back to the original frame. We now need to relate η to something tangible like velocity.



$$\begin{pmatrix} t \\ x \end{pmatrix} \cdot u_+ = \frac{t+x}{\sqrt{2}}$$

$$\begin{pmatrix} t \\ x \end{pmatrix} \cdot u_- = \frac{t-x}{\sqrt{2}}$$

$\therefore$ Any event vector can be decomposed in this space as: $\begin{pmatrix} t \\ x \end{pmatrix} = \left(\frac{t+x}{\sqrt{2}}\right) u_+ + \left(\frac{t-x}{\sqrt{2}}\right) u_-$

This makes applying a lorentz tranformation to this vector, much easier.

$$\Rightarrow \Lambda \begin{pmatrix} t \\ x \end{pmatrix} = \left(\frac{t+x}{\sqrt{2}}\right) e^{-\eta} u_+ + \left(\frac{t-x}{\sqrt{2}}\right) e^{\eta} u_-$$

$$= \begin{pmatrix} t' \\ x' \end{pmatrix} = \left(\frac{t'+x'}{\sqrt{2}}\right) u_+ + \left(\frac{t'-x'}{\sqrt{2}}\right) u_-$$

$$\boxed{\begin{aligned} t'+x' &= e^{-\eta}(t+x) \\ t'-x' &= e^{\eta}(t-x) \end{aligned}}$$

t and x, individually, get rotated.
$t \pm x$, together, get scaled

However, we see an invariant.

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Note the following: $(t'+x')(t'-x') = t'^2 - x'^2$

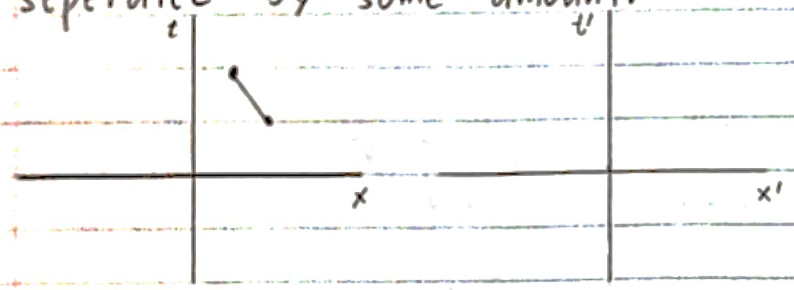
But remember the transformations:

$$= (t+x)(t-x) = t^2 - x^2$$

Therefore, we can conclude: $\boxed{t'^2 - x'^2 = t^2 - x^2}$

This is an invariant of the lorentz transformation, aka the "space-time interval".

Driving home the point, if you have 2 events seperated by some amount:



The same 2 events will have 2 dif coords. But both observers will agree on the quantity $t^2 - x^2$. It also works the other way.

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Conversion back to Standard Basis

$$t'+x' = e^{-\eta}(t+x)$$
$$t'-x' = e^{\eta}(t-x)$$

Add: $t' = \frac{1}{2}\left[e^{-\eta}(t+x) + e^{\eta}(t-x)\right]$
$$= \tfrac{1}{2}\left[(e^{\eta}+e^{-\eta})t - (e^{\eta}-e^{-\eta})x\right]$$
$$= \cosh\eta\, t - \sinh\eta\, x$$

Subtract: $x' = -\sinh\eta\, t + \cosh\eta\, x$

Matrix form: $$\begin{pmatrix} t' \\ x' \end{pmatrix} = \underbrace{\begin{pmatrix} \cosh\eta & -\sinh\eta \\ -\sinh\eta & \cosh\eta \end{pmatrix}}_{\Lambda(\eta)} \begin{pmatrix} t \\ x \end{pmatrix}$$

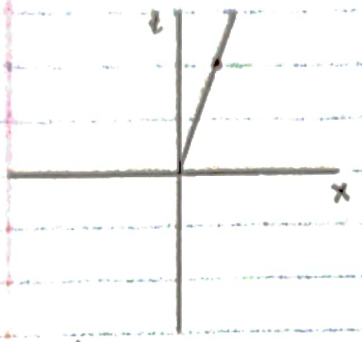
η = "rapidity", related to v.
$\eta = 0$ when $v = 0$

The nice thing about rapidities is that they add, i.e. two consecutive lorenz transformations:

$$\Lambda(\eta_1)\Lambda(\eta_2) = \Lambda(\eta_1+\eta_2)$$

Referring to "nice" variables: $t'-x' = e^{\eta_2}(t+x)$
$$t''-x'' = e^{\eta_1}(t'-x')$$
$$= e^{\eta_1}e^{\eta_2}(t+x)$$
$$= e^{\eta_1+\eta_2}(t+x)$$

But we still need to relate η to v. Use the fact that the origin of the primed frame moves with velocity v.



Relating η to v:

$$\begin{pmatrix} t' \\ 0 \end{pmatrix} = \Lambda \begin{pmatrix} t \\ vt \end{pmatrix} = \begin{pmatrix} \cosh\eta & -\sinh\eta \\ -\sinh\eta & \cosh\eta \end{pmatrix} \begin{pmatrix} 1 \\ v \end{pmatrix} \cancel{t}$$

$$-\sinh\eta + \cosh\eta\, v = 0$$

$$\sinh\eta = \cosh\eta\, v \Rightarrow \boxed{\tanh\eta = v}$$

$$\frac{e^{\eta} - e^{-\eta}}{e^{\eta} + e^{-\eta}} \to \frac{e^{\eta}}{e^{\eta}} = 1$$

If the rapidity is very large, then 1 is the speed of light aka our units for c is 1. This means for the limit as $\eta \to \infty$, the prime frame approaches the speed of light.

Using above: $\Lambda = \cosh\eta(v)\begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}$

$\Lambda^{-1} = \cosh\eta(-v)\begin{pmatrix} 1 & v \\ v & 1 \end{pmatrix}$

$$\cosh\eta(v)\cosh\eta(-v)\begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}\begin{pmatrix} 1 & v \\ v & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \cosh\eta(v)\cosh\eta(-v)\begin{pmatrix} 1-v^2 & 0 \\ 0 & 1-v^2 \end{pmatrix}$$

$$\cosh\eta(v)\cosh\eta(-v) = \frac{1}{1-v^2}$$

$$t' - x' = e^{\eta(v)}(t+x)$$

$$t + x = e^{\eta(-v)}(t'-x')$$

$$\Rightarrow \eta(-v) = -\eta(v)$$

$$\Rightarrow \cosh\eta(-v) = \cosh\eta(v)$$

$$\cosh\eta(v)^2 = \frac{1}{1-v^2}$$

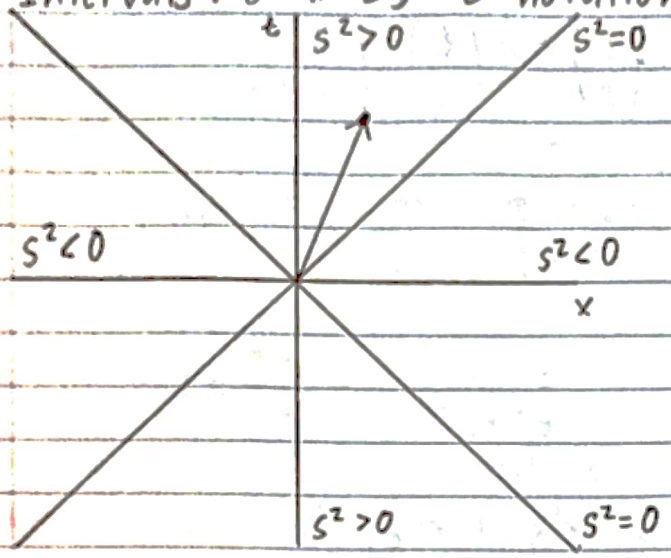
$$\cosh\eta(v) = \frac{1}{\sqrt{1-v^2}} = \gamma$$

$$\boxed{\Lambda = \gamma\begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}, \quad \gamma = \frac{1}{\sqrt{1-v^2}} \qquad \Lambda = \begin{pmatrix} \cosh\eta & -\sinh\eta \\ -\sinh\eta & \cosh\eta \end{pmatrix}, \quad \tanh\eta = v}$$

This relabels points in space-time, NOT move them. Hilroy

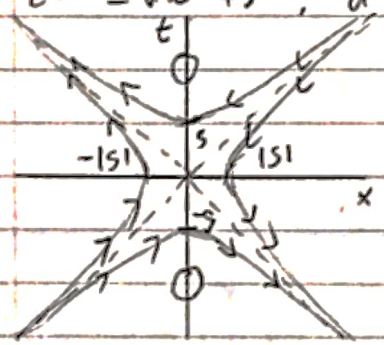
Understanding the Transformations

Intervals: $t^2 - x^2 = s^2$ ← notation



When $s^2 > 0 \Rightarrow$ timelike
$s^2 < 0 \Rightarrow$ space-like
$s^2 = 0 \Rightarrow$ null

After a lorentz transformation, s^2's value does not change. T and x can change. This means in the timelike region, we can say $t = \pm\sqrt{x^2+s^2}$, a hyperbola, same with spacelike.



timelike: $t = \pm\sqrt{x^2+s^2}$
spacelike: $x = \pm\sqrt{t^2-s^2}$
$\uparrow$
$s^2<0$

More importantly points in the spacelike region cannot move; you can never take a point in the timelike region into the spacelike region using a lorenz transformation, vice-versa. Null directions stay on the null directions.
↳ A point never changes its category.

Two events seperated by spacelike vector cannot be causally related. We can't say something in the origin causes a change in the hyperbola.

Everything that is contained in the lorenz transformation is just a question of backing it up.

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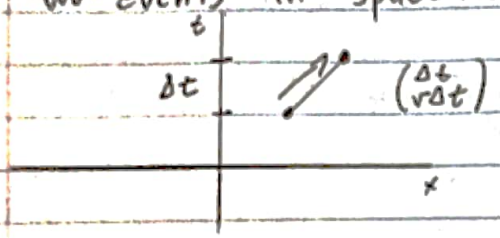
Time Dilation

Two events in spacetime seperated by Δt:



What is the time elapsed by a clock that moves along the trajectory? (Find $\Delta t'$)

Note that if the clock is moving along the trajectory, no space is elapsed - it sits in the origin of its own frame.

$$\begin{pmatrix} \Delta t' \\ 0 \end{pmatrix} = \gamma \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix} \begin{pmatrix} \Delta t \\ v\Delta t \end{pmatrix}$$

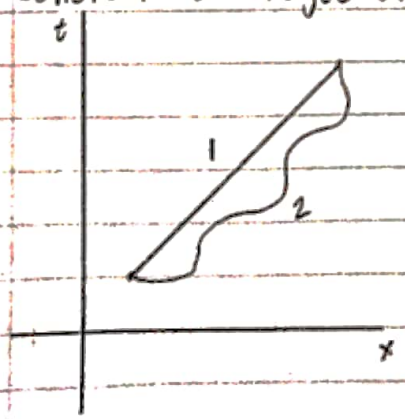
$$\Delta t' = \gamma(\Delta t - v^2 \Delta t)$$

$$\Rightarrow \Delta t' = \sqrt{1 - v^2}\, \Delta t$$

$$\boxed{\Delta t' = \frac{\Delta t}{\gamma}} \longrightarrow \text{Because } \gamma > 1,\ \Delta t' < \Delta t$$

It is customery to call the time elapsed by a moving clock "proper time", denoted $\Delta \tau$ instead of $\Delta t'$. Therefore: $\boxed{\Delta \tau = \frac{\Delta t}{\gamma}}$

Consider 2 trajectories in spacetime:

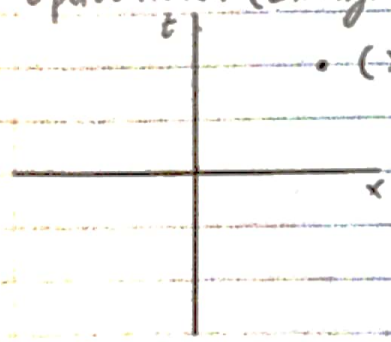


Path 1 will have more proper time than path 2, Why?

Index Notation

This is the workhorse of calculations in QFT.

Spacetime! (Imagine in 3D)



• $(x^0 = t, x^1 = x, x^2 = y, x^3 = z)$

Not squared or cubed. Coords are collected into an object, x^μ

$\mu = 0, 1, 2, 3$
- 0 = time
- 1 = x
- 2 = y
- 3 = z

Different observers will assign different coords to the same event.

The event above will have different coords, We'll call them x'^μ. This will equal some matrix.

$$x'^\mu = \sum_\nu \Lambda^{\mu\nu} x^\nu$$

← inner indices match, just matrix mult. and simplification of notation

$$= \sum_\nu \Lambda^\mu{}_\nu x^\nu$$

$$= \Lambda^\mu{}_\nu x^\nu$$

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu$$

Interval: $s^2 = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2$

$g_{\mu\nu}$ ← $g_{00} = 1$, $g_{11} = g_{22} = g_{33} = -1$

$$g = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$$

$\therefore s^2 = x^\mu x^\nu g_{\mu\nu}$ ← Review notation

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$$S^2 = x^\mu x^\nu g_{\mu\nu}$$

$$x \cdot x = x^2 \equiv x^\mu x^\nu g_{\mu\nu}$$

$$x_\mu \equiv g_{\mu\nu} x^\nu \leftarrow$$

$$x^2 = x^\mu x^\nu g_{\mu\mu} = x^\mu x_\mu$$

↳ Inverse metric $= g^{\mu\nu}$

↓

By def: $g^{\mu\nu} g_{\nu\sigma} = \delta^\mu_\sigma = \begin{cases} 0 & \text{if } \mu \neq \nu \\ 1 & \text{if } \mu = \nu \end{cases}$

Example: $x_\mu = g_{\mu\nu} x^\nu$

$$g^{\sigma\mu} x_\mu = g^{\sigma\mu} g_{\mu\nu} x^\nu$$

$$= \delta^\sigma_\nu x^\nu = x^\sigma$$

$$\boxed{x^\mu = g^{\mu\nu} x_\nu} \rightarrow \text{Done!}$$

Identities:

$$(x+y)^2 = x^2 + y^2 + 2x \cdot y = (x' + y')^2 \leftarrow \text{invariant}$$

$$= x'^2 + y'^2 + 2x' \cdot y'$$

$$x \cdot y = x' \cdot y'$$

$$x^\mu y^\nu g_{\mu\nu} = x'^\mu y'^\nu g_{\mu\nu}$$

Using transformation, $x'^\mu = \Lambda^\mu_\alpha x^\alpha$

$$= (\Lambda^\mu_\alpha x^\alpha)(\Lambda^\nu_\beta y^\beta) g_{\mu\nu} \leftarrow \text{These are numbers individual components}$$

$$= x^\alpha y^\beta (\Lambda^\mu_\alpha \Lambda^\nu_\beta g_{\mu\nu})$$

$= x^\alpha y^\beta (g_{\alpha\beta})$, choose $x^\alpha = (1,0,0,0)$ if you want

$$\boxed{\Lambda^\mu_\alpha \Lambda^\nu_\beta g_{\mu\nu} = g_{\alpha\beta}}$$

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$$\Lambda^{\mu}_{\alpha}(\Lambda^{-1})^{\alpha}_{\sigma}\Lambda^{\nu}_{\beta}g_{\mu\nu} = (\Lambda^{-1})^{\alpha}_{\sigma}g_{\alpha\beta}$$

$$\delta^{\mu}_{\sigma}\Lambda^{\nu}_{\beta}g_{\mu\nu} = (\Lambda^{-1})^{\alpha}_{\sigma}g_{\alpha\beta}$$

$$\Lambda^{\nu}_{\beta}g_{\sigma\nu} = (\Lambda^{-1})^{\alpha}_{\sigma}g_{\alpha\beta}$$

$$\boxed{\Lambda^{\alpha}_{\beta}g_{\nu\alpha} = (\Lambda^{-1})^{\alpha}_{\nu}g_{\mu\alpha}}$$

$$\begin{aligned} x'_{\mu} &= g_{\mu\nu}x'^{\nu} \\ &= g_{\mu\nu}\Lambda^{\nu}_{\alpha}x^{\alpha} \\ &= (\Lambda^{-1})^{\nu}_{\mu}g_{\nu\alpha}x^{\alpha} \\ &= (\Lambda^{-1})^{\nu}_{\mu}x_{\nu} \end{aligned}$$

$$\boxed{\begin{aligned} x'_{\mu} &= x_{\nu}(\Lambda^{-1})^{\nu}_{\mu} \\ x'^{\mu} &= \Lambda^{\mu}_{\nu}x^{\nu} \end{aligned}}$$

$$x_{\mu} = g_{\mu\nu}x^{\nu}$$

$$\begin{aligned} x'^{\mu}y'_{\mu} &= [\Lambda^{\mu}_{\nu}x^{\nu}][(\Lambda^{-1})^{\sigma}_{\mu}x_{\sigma}] \\ &= \Lambda^{\mu}_{\nu}(\Lambda^{-1})^{\sigma}x^{\nu}y_{\sigma} \\ &= \delta^{\sigma}_{\nu}x^{\nu}y_{\sigma} \end{aligned}$$

$$\boxed{x'^{\mu}y'_{\mu} = x^{\sigma}y_{\sigma}} \leftarrow$$

T'^{μ}_{μ}

Next Time:
- relativistic classical mechanics
- relativistic QM of free particles

Partial derivative:

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu}$$

$$\partial'_\mu = \frac{\partial}{\partial x'^\mu} = \frac{\partial x^\nu}{\partial x'^\mu}\partial_\nu$$

$$x^\nu = (\Lambda^{-1})^\nu{}_\mu x^\mu$$

$$\frac{\partial x^\nu}{\partial x'^\mu} = (\Lambda^{-1})^\nu{}_\mu$$

$$\Rightarrow \boxed{\partial'_\mu = (\Lambda^{-1})^\nu{}_\mu \partial_\nu}$$

$$x'_\mu = x_\nu (\Lambda^{-1})^\nu{}_m$$

$\Rightarrow \rho\, dv$ = scalar

$\rho\, dv\, dx^\mu$ = 4-scalar

$$= \underbrace{\rho\, dv\, dt}_{\underbrace{dx^0 dx^1 dx^2 dx^3}} \frac{dx^\mu}{dt} = \rho\, d^4x \frac{dx^\mu}{dt}$$

$$\int d^4x\, f(x) = \int d^4x' \left|\frac{\partial x}{\partial x'}\right| f(x(x'))$$

$$M^\mu{}_\nu = \frac{\partial x^\mu}{\partial x'^\nu} = (\Lambda^{-1})^\mu{}_\nu$$

$$d^4x' = d^4x$$

$$\Rightarrow \rho \frac{dx^\mu}{dt} = j^\mu, \quad j^0 = \rho \frac{dt}{dt} = \rho$$

$$j^i = \rho \frac{dx^i}{dt} = \rho v^i$$

$$j = \begin{pmatrix} \rho \\ \vec{j} \end{pmatrix}$$

$$\frac{d\rho}{dt} + \vec{\nabla}\cdot\vec{j} = 0$$

$$\partial_0 j^0 + \sum_i \partial_i j^i = 0$$

$$\boxed{\partial_\mu j^\mu = 0}$$